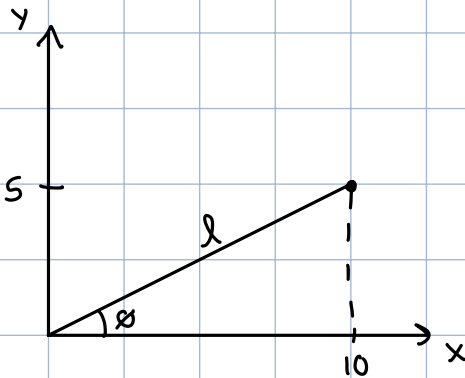


Target point: (10, 5, 5)

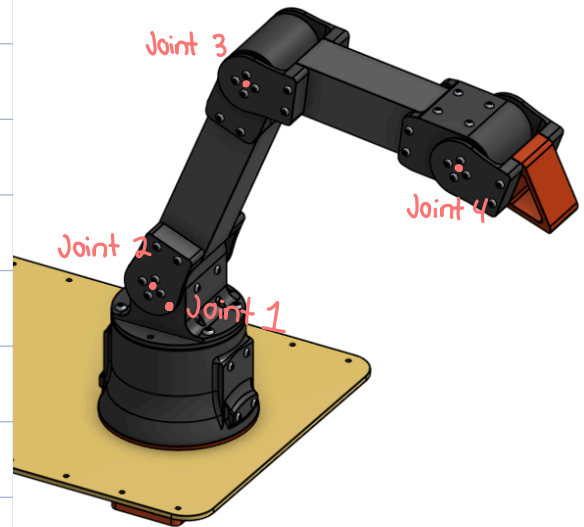
Base angle (Joint 1)



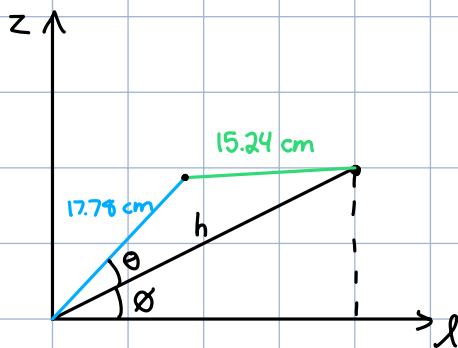
will use atan²
for negatives

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$l = \sqrt{x^2 + y^2}$$



Joint 2 and 3



$$\phi = \tan^{-1}\left(\frac{z}{l}\right)$$

$$h = \sqrt{z^2 + l^2}$$

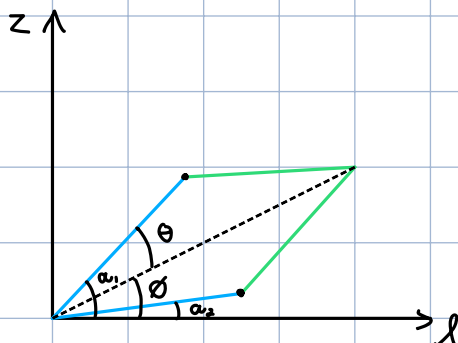
Joint 4

Approaching horizontally for now
so wrist angle is fixed

$$15.24^2 = 17.78^2 + h^2 - 2 \cdot 15.24 h \cos \theta$$

$$2 \cdot 15.24 h \cos \theta = 17.78^2 - 15.24^2 + h^2$$

$$\theta = \cos^{-1}\left(\frac{17.78^2 - 15.24^2 + h^2}{2 \cdot 15.24 h}\right)$$



$$\alpha_1 = \phi + \theta \quad (\text{angle for Joint 2})$$

$$\alpha_2 = \phi - \theta \quad (\text{angle for Joint 3})$$